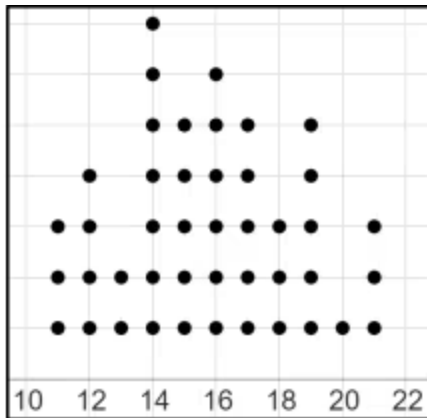




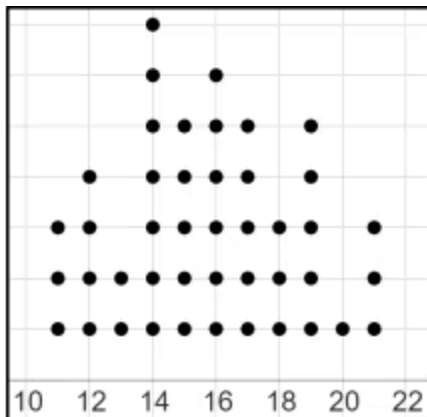
Changing a data point

- 1 One data point on this dot plot was incorrectly typed in. After the data point is modified and corrected, Andeep notices the mode has changed. Which of these statements must be true? Tick 2 correct answers



- ☒ The incorrect data point is 14.
- ☐ The incorrect data point is 16.
- ☐ The incorrect data point is 20.
- ☐ The data point must be corrected to 14.
- ☒ The data point must be corrected to 16.

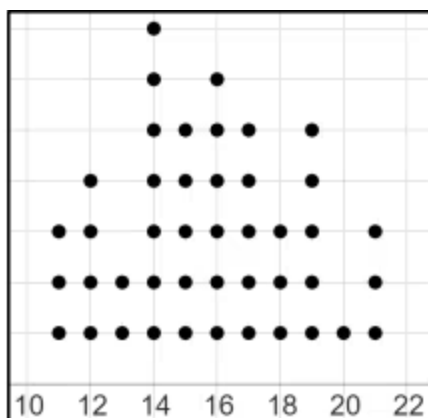
- 2 The mean of this dot plot is currently 15.75. A new data point is collected and added to the data set. Laura re-calculates the mean and notices it has decreased. What could the new data point be? Tick 2 correct answers



- ☒ 9
- ☒ 15
- ☐ 15.75
- ☐ 16
- ☐ 22



- 3 The mean of this dot plot is 15.75. A data point is no longer relevant and removed from the data set. Sam re-calculates the mean and notices it has decreased. What could the new data point be? Tick 2 correct answers



- ☐ 9
- ☐ 15
- ☐ 15.75
- ☒ 16
- ☒ 22

- 4 A data point is removed from this dataset, as it is no longer relevant to an investigation. Which of the following are possible outcomes after the data point is removed? Tick 2 correct answers

-6	-5	-3	-3	1	4	5	7
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- ☐ The data set may become bimodal.
- ☒ The mode of the data set may stay the same.
- ☐ The mode of the data set may change to a different value.
- ☒ There may be no mode in the data set.



- 5 Match each possible change to the data set to the correct description of its impact on the median. Write the correct letter in each box

4	4	4	4	9	9	9	9
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a	A data point of 2 is added
b	A data point of 4 is removed
c	A data point of 6.5 is added
d	4 is modified in value to a 7

<i>d</i>	The median becomes 8.
<i>b</i>	The median becomes 9.
<i>c</i>	The median stays the same.
<i>a</i>	The median becomes 4.



- 6 Match each possible change to the data set to the correct description of its impact on the mean. Write the correct letter in each box

4	4	4	4	9	9	9	9
---	---	---	---	---	---	---	---

a	4 is modified in value to a 20
b	9 is modified in value to a 1
c	A data point of 20 is added
d	A data point of 6.5 is added
e	All the 9s are removed

c	The mean increases to 8.
e	The mean decreases to 4.
a	The mean increases by 2.
d	The mean stays the same.
b	The mean decreases by 1.

